A detailed account of Case Study 1: Women and other minority communities in STEM

How the study proceeded, from the first count to the final findings

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This account follows the case study in the order in which each step depended on the one before it. For every step it says what we saw, what we did, how we checked it, what we found, whether it explained anything, what we learnt or corrected, and what it made us do next.

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1. The whole study on one page

Stage	Question	What we did	What we found	What it led to
1 Count	Who is in the room at IISER Kolkata, and is IISER Kolkata unusual?	Read every NIRF submission for IISER Kolkata (2019–2026), then for the seven IISERs and IISc (44 PDFs).	Women are 27.2% of five-year students and 14.4% of regular faculty at IISER Kolkata. IISER Kolkata and Pune are lowest for students (27.2%, 27.6%); Tirupati and Thiruvananthapuram are about 52%. IISER Kolkata's faculty share has sat at 13.8–14.4% in all eight lists.	Is this our campus, or the national system?
2 Place	How does this compare with India and with the Institutes of National Importance (INIs)?	Compared our findings with AISHE reports 2019-20 to 2023-24.	Women are 49.7% of students and 44.9% of teachers in India, but 26.5% of INI undergraduates and 22.1% of INI teachers. The gap is a feature of the INIs and of faculty, not only of IISER Kolkata.	Who else is missing, and at which step?
3 Widen	Do SC, ST and OBC students and faculty show the same drop? Where do women drop off?	AISHE by level, by category and by institution type; NIRF student category columns; IISER Kolkata faculty counts from the Dean of Faculty Affairs	Nationally SC and ST shares fall most from postgraduate level to the PhD. At the IISERs students are close to the 49.5% reserved share (47.6% pooled), but IISER Kolkata's faculty are 16.0% (21 of 131) from these groups. Women lead only 8.5% of INIs.	The numbers say how few, not why.
4 Ask	Why?	Eight interviews: the two deans and six women faculty.	Mostly social and structural: who reaches the top entrance ranks, the two-body problem, location, hostels for women PhD students, slow hiring since about 2012, few women applicants, no credit for career breaks.	Four ideas to test, and a request for grade data.
5 Test	Do home state, faculty size, rank and preference, or grades explain it?	We tested each idea and wrote detailed reports of our findings: Report 3 (Section 5) for home state, Report 4.1 for faculty size, Reports 10, 11 and 17 for grades, and Report 13 for rank and preference	Home state: no. Faculty size: no. Grades: women **DO NOT** do worse overall; subject-wise gaps line up with who takes the department across subjects, but not conclusively. Rank and preference: a pattern (Spearman 0.82 across the seven IISERs, 0.88 with IISc), now written up as its own report.	Does the loss start before IISER?

Stage	Question	What we did	What we found	What it led to
6 Look earlier	Where do girls stop studying?	Field survey of 91 women in Ghoragacha and Mohanpur (Report 12; 90 analysed, as one respondent was a minor), the student survey and a follow-up with Dr. Rumi De.	Marriage and poverty ended schooling; the decision was usually someone else's; younger women have studied much further than older women; many still want to learn.	A collaborative education drive with Ek Pehal for kids in the nearby villages (done and ongoing), preparation of free and standardised educational content in English, Hindi and Bengali, education visits to schools (apart from the ones where we went for regular weekly sessions for Ek Pehal) (done), talking with experts (done), a plan to talk to local NGOs on providing skill training on sewing, etc. (planned)

2. The reports

Sl. No.	What it is	Built from	Main result
2	IISER Kolkata: what the NIRF data show, 2017-18 to 2024-25	NIRF "Overall" submissions of IISER Kolkata, 2019–2026 (8 files)	Women 27.2% of five-year students, 14.4% of regular faculty (18 of 125); flat since 2019
3	Women and other minority communities in IISERs and IISc: enrolment	44 NIRF PDFs, 79 programme rows	Range 27.2–51.8%; no general decline; seats grew but share did not follow; home state not an explanation (added as a later section after interviews with Dr. Bhadra and Dr. De)
4	Women among IISER and IISc faculty	42 of 44 datasets, 7,400 entries	Range 12.0–23.2%; IISER Kolkata flat; other IISERs' recent hires are more often women, IISER Kolkata's are not
4.1	Faculty size and IISER Kolkata's low share	Latest NIRF faculty lists	IISER Kolkata is third-largest of seven (125); difference from the other six not distinguishable from chance ($p = 0.21$)
5	Women among students and teachers in India	AISHE 2019-20, 2021-22, 2022-23, 2023-24	49.7% of students, 44.9% of teachers, 31.3% of professors, 22.1% at INIs
2.1	IISER Kolkata against other IISERs, INIs and India (one page summary)	NIRF and AISHE 2023-24	IISER Kolkata lowest of seven IISERs for students; faculty below INI average
6	Women in top administrative posts	Published surveys, press, IIT directors' list	152 of 1,352 institutions (11.24%); INIs 14 of 164; no IIT has a woman in a top administrative role
7	Who reaches the PhD? SC, ST, OBC and women	AISHE (Tables 6, 11, 13, 27, 33a, 43, 45, 50)	SC 15.6% at UG to 10.0% at PhD; ST 6.3% to 4.1%; women at parity nationally but not in West Bengal or at INIs
8	SC, ST and OBC students at the IISERs (and IISER Kolkata faculty)	NIRF student tables; DoFA office counts	47.6% pooled (4,418 of 9,276); IISER Kolkata faculty 21 of 131 (16.0%)
9	SC, ST and OBC faculty in India	AISHE (Tables 11, 14, 21, 27, 30)	INI faculty 26.1% against 46.3% of INI students (ratio 0.56)
10	Gender and CGPA across batches (for IISER Kolkata)	DoAA: CGPA graphs for 8 batches; subject GPA for 6	Mixed in the first two years; women ahead later (semester 10: +0.14)
11	Subject-wise GPA by gender (for IISER Kolkata)	DoAA table, 6 batches	Humanities the only consistent gap (+0.16). The gaps in other subjects are not conclusive.
17	Department take-up and grades (for IISER Kolkata)	DoAA men-per-woman ratios, 8 batches; the previous two reports	Across subjects rank correlation 0.90 (exact $p = 0.08$); within a department −0.16 per 10 points
13	Rank and preference: are the most preferred IISERs the ones with the fewest women?	NIRF student and faculty tables; the preference order quoted in the interviews	Women's share rises as an institute sits lower in the NIRF order: $\rho = 0.82$ (exact $p = 0.03$), 0.88 with IISc added ($p = 0.007$); preference order 0.86 ($p = 0.02$); faculty shows no pattern (−0.32)

Sl. No.	What it is	Built from	Main result
12	Why did schooling stop? Village survey	91 women interviewed (90 analysed, as one was a minor), from Ghoragacha and Mohanpur	25% never attended school; reasons: marriage 59%, cost 55%
15	Report for IISER student councils	All the above reports	Summary and requests for data

3. What each interview taught us

Eight interviews, with two deans and six women faculty. For each: what we learnt, and what it changed in the study. A summary is provided below. The entire interviews and interview questions are provided as "Interviews and surveys".

Who	What we learnt	What it changed
Dean of Faculty Affairs (Dr. Rahul Banerjee)	**Hiring follows government norms;** there is no reservation for women, but the institute set its own target (4 of 8 in the latest joint recruitment, 3 of 7 after one left; about 8 of 40+ recruits since 2013). Few women apply (about 20 of nearly 200 in one Chemistry search). Location and spouses' jobs cost the institute candidates. Hostels for women PhD students are being built. For SC/ST/OBC faculty the quota was only recommended before 2019 and mandated after.	Gave us the faculty and reserved-category counts, and led us to interview the women faculty. Helped us understand why the faculty counts from the SC, ST and OBC communities are so low.
Dr. Anindita Bhadra	Hostel conditions may deter parents; the share of Kerala students at IISER Kolkata fell once the two southern IISERs opened; location and the two-body problem; the biology–physics gap is a global, mostly social pattern; the environment for women in STEM is harder than we had believed.	Suggested the home-state test and, later, nearby states (untestable). Led to the women-in-STEM talk series idea.
Dr. Malancha Ta	Two-body problem and a dearth of suitable employment opportunities for spouses in the nearby regions; women in leadership matter more than role models; girls do slightly better than boys in school boards, so something changes later; asked for applicants by gender (an RTI would be needed).	Sharpened the leadership point and the data gaps.
Dr. Anuradha Bhat	Female representation is almost the inverse of institute ranking and preference; the top ranks go to students with coaching and family backing, which girls are often held back from; the small faculty size might make percentages unreliable.	Changed the direction of the study: the gap probably begins well before IISER. Led to Report 4.1 and to the rank and preference analysis (Report 13).
Dr. Neelanjana Sengupta	Few women apply while standards stay high; the leaky pipeline and social expectations; two-body problem; tokenism and patronising attitudes; biology is broad and welcoming; the institute informally prefers the woman when two candidates are equal.	Confirmed the pipeline picture; added tokenism.
Dr. Koel Das	Hiring was slow after 2012, so women faculty are mostly senior; no general recruitment until about 2018-19; spouse hiring should be visible; the US "nerd" explanation may not apply in India; more women are needed in decision-making; hostels for PhD students.	Explained why IISER Kolkata's women are concentrated in senior ranks; gave concrete suggestions.
Dean of Academic Affairs (Dr. Rangeet Bhattacharyya)	Women do as well as or better than men once admitted, so capability is not the reason; women choose different departments; a study cited by Prof. Godbole found that question framing changes marks; women drop off after marriage and children; the SC/ST/OBC rule change in 2019.	Gave the CGPA and subject data behind Reports 10, 11 and 17, and led to Report 7.

Who	What we learnt	What it changed
Dr. Rumi De	Admission is rank-wise, the top of the list is dominated by boys, and the cause is guidance, coaching and family investment, not ability; the south is seen as safer; nearby states; career breaks are not credited in hiring.	Explained the rank and preference pattern. A second conversation suggested NGOs and skills training for women in the settlements.

4. What each idea led to, and what filled what

4.1 The explanations, as hypotheses

Every reason we heard is listed here once, with who raised it, what we did with it, and where it stands. "Not testable" means the data that would test it are not public or were not given to us.

Idea	Raised by	What we did	Where it stands
Women do worse once admitted (capability)	The first hypothesis people form (DoAA note)	CGPA and subject GPA: Reports 10, 11	Not supported overall. Women are level or ahead later; subject gaps are small. Selection could hide a real difference.
Home state	The early NIRF look; Dr. Bhadra	Report 3 (Section 5)	Not supported (seven to eight institutes, wide intervals).
Neighbouring states	Dr. Bhadra, Dr. De	Nothing possible: NIRF has only home state	Open. Needs state-of-origin data from the institutes.
A small faculty makes percentages unreliable	Dr. Bhat	Report 4.1	Not the explanation; narrowed our claim ($p = 0.21$ against the other six IISERs).
Who reaches the top entrance ranks; coaching and family investment	Dr. Bhat, Dr. De, Dr. Sengupta	Report 13	Supported as a pattern: $\rho =$ 0.82 (NIRF order), 0.86 (preference order), 0.88 with IISc added; **mechanism not testable** without applicant and rank data by gender.
Two-body problem and location	DoFA, DoAA, Dr. Das, Dr. Bhadra, Dr. Ta, Dr. Sengupta, Dr. Bhat, Dr. De	None possible	Raised by all of the eight. The DoFA lost a candidate over it. Not testable.
Few women apply	DoFA, Dr. Das, Dr. Sengupta	The same reason may not apply to other institutes. There is no way of testing this without an RTI.	Applicants not stored by gender. Open.
Slow hiring since about 2012; a senior-heavy faculty	DoFA, Dr. Das	Reports 2, 4	Consistent with the data (18 assistant professors; cohort pattern), not proven.
Career breaks are not credited	Dr. De	None	Not tested.
Few women in leadership	Dr. Das, Dr. Ta	Report 6	8.5% of INIs are headed by women; the pool is larger than the posts.
Role models	All; views differ	Student survey only (small scale survey: not a random sample)	Mixed. About half of surveyed women named them; some faculty put women in power ahead of role models.
Test and question framing	DoAA	None; cited a published study	Not tested here.

Idea	Raised by	What we did	Where it stands
Social conditioning away from the "hard" sciences	Dr. Bhadra, Dr. Ta, Dr. Sengupta, Dr. Bhat, Dr. Das, Dr. De	Not directly testable; Reports 11 and 17 show the pattern by department	Consistent with the data. A cause the data cannot confirm.
Marriage and family pressure later in the pipeline	DoAA, Dr. De, Dr. Das	Though widely acknowledged as a plausible reason, we have not directly tested it. Report 7 (national); Report 12 (local). However, all the interviewees confirmed that this is a major reason. Given the opinions from all experts, we also come to the same conclusion (despite not directly testing for it)	Report 7 finds the PhD drop in some places (West Bengal, INIs); Report 12 finds marriage ending schooling for younger women. Not tested at IISER Kolkata.
Tokenism and patronising attitudes	Dr. Sengupta, Dr. Koel Das, Dr. Malancha Ta	None. However, we do acknowledge it as a legitimate concern given the expert opinions we heard.	Reported experience

4.2 What filled what

Each report left something it could not see. This table shows what filled each gap and what is still empty.

What we could not see	What filled it	Still empty
Why IISER Kolkata is low (the reports describe, they do not explain)	The eight interviews	Which reasons carry the weight is not known
Faculty by category at IISER Kolkata (NIRF has no column)	DoFA office counts: 21 of 131 (Reports 8, 9)	Same figures for the other IISERs
PhD students and postdocs by gender (NIRF gives PhD totals only)	AISHE nationally (Report 7); a figure of about 32% for IISER Kolkata's PhD share (about 27% at BS-MS)	PhD and postdocs by gender and category at any IISER
IISERs inside the INI category (AISHE reports INIs together)	NIRF for the IISERs themselves (Reports 3, 4, 8)	Whether the IISERs differ from the IITs, NITs and AIIMS in AISHE
Whether women do worse	The CGPA graphs and subject GPA given by the dean (Reports 10, 11)	Counts, standard deviations, single-semester SGPA
Whether grades line up with who takes the department	The data given by the dean on men-per-woman ratios (Report 17)	Counts behind the ratios; grades after the major; admission ranks by gender
Faculty by department at IISER Kolkata (NIRF lists no field)	Department counts from the DoFA office	Department counts at other IISERs

What we could not see	What filled it	Still empty
Home state and nearby states	NIRF home-state share (Report 3, Section 5)	State of origin of students
Is a small faculty the problem	Report 4.1	Nothing further
Who leads	Report 6	Anyone below director
Where the loss starts before IISER	Village survey (Report 12), Dr. De's follow-up	Girls today; women not at home when we visited; school records
Applicants, offers and joiners by gender, category and route (faculty and students)	Nothing: not stored or not public (entrance test; an RTI would be needed)	All of it
What students say	A small student survey to understand the general trends among the students	A larger, more comprehensive survey

5. Corrections, revisions and things to look at

5.1 Corrections the study made on its way

These are visible in the final texts. Each one changed what we said or how we said it.

What	Where	What happened
Berhampur's category share fell 8.4 points while its student total grew by 183	Reports 8, 15	Treated as a probable reporting change; shown with and without (47.6% against 48.6% pooled); Berhampur invited to correct it.
"Economically backward" column	Reports 3, 8	Looked at, found swinging implausibly (IISER Kolkata 15%, 12%, 28%, 27%, 35%, 9%, 9%, 10%), dropped as unusable.
One-year category moves above 3 points	Report 8, App. B	Listed (seven institute-years) and treated as possible reporting changes.
IISER Kolkata faculty "lower than the other IISERs"	Reports 2, 4 → 4.1	Narrowed: not distinguishable from chance against the other six; supported against the INI average and India.
Home-state hypothesis	Report 3 (Section 5)	Tested three ways; not supported; neighbouring states untestable.
Director counted in NIRF's 125	Report 8	The institute's 131 excludes the director; reconciled (124 + 7 = 131)
PhD count jump of 47%	Report 7, 9	Reason not found; all comparisons use shares; flagged.
Women vice-chancellors figure	Report 6	The article prints 0.024%; the figure is about 1.8%; corrected in the text.
Report 10's early gap	Report 10	The headline average (+0.07 at semester 4) is pulled by two batches; the report gives the figure without them.
Test counts	Reports 11, 17	Both state the number of tests and that none would survive strict correction for multiple testing.
Pooled department-by-batch result	Report 17	Reported as 0.06 and explained as two opposite effects cancelling, not hidden.
Missing years and files	Reports 3, 4, 5, 7	AISHE 2020-21; Mohali 2020 faculty; Mohali 2026; Thiruvananthapuram 2019; IISc 2018 faculty. Stated, and trends skip them.

6. What we finally found

The numbers below are the final ones from the reports, grouped by what they say, with what each can and cannot show.

Women as students

- Women are 27.2% of five-year UG students at IISER Kolkata (286 of 1,050; 2024-25), up from 24.2% in 2017-18, almost all in one step in 2020-21. IISER Kolkata and Pune (27.6%) are lowest of the seven IISERs; Tirupati (51.8%) and Thiruvananthapuram (51.6%) are about twice as high. The order has been stable. Women's share rose at Mohali, Thiruvananthapuram and IISER Kolkata and slipped at Pune (−2.2 points since 2017-18); Berhampur and Tirupati have too few years to judge.
- Seat expansion did not by itself raise the share: what mattered was who filled the new places (women were 77% of the growth at Mohali, 23% at Pune). Women's share is higher in PG than in the five-year programme at Pune and IISER Kolkata, a pointer to entry at UG level.
- Nationally, women are about half of students at every level (49.5% UG), but 26.5% of UG students at INIs. IISERs sit in between (40.4% pooled).

Women as faculty

- Women are 14.4% of regular faculty at IISER Kolkata in the NIRF 2026 list (18 of 125), flat at 13.8–14.4% in all eight lists since 2019; with the seven newest appointments (3 women) the DoFA's counts imply about 16.0% (21 of 131). The seven IISERs pooled are 18.8%; INIs 22.1%; India 44.9%. IISER Kolkata's share is below the INI average ($p \approx 0.04$), not distinguishable from the other six IISERs ($p = 0.21$), and cannot be explained by its size (125 faculty, third-largest).
- At the other six IISERs women are a larger share of more recent joiners (14%, 18%, 22%, 26%); at IISER Kolkata they were not until the latest cycle. IISER Kolkata has only 18 assistant professors. Its women are concentrated in senior ranks, because most joined around 2010–2012 and hiring has been slow since.
- Women are a smaller share of faculty than of students at every IISER: the fall from UG to faculty is 21.6 points at the IISERs against 4.4 at INIs and 4.6 for India. The stages are **different groups of people**, so **this is NOT a loss of the same women**.

Women in the pipeline and in leadership

- Nationally women reach parity at the PhD (50.1%) but not in West Bengal (51.3% at UG, 39.8% at PhD), not in engineering (37%) and not at INIs (34.6%, 14.0 points below all universities). SC women are the thinnest group at the PhD (44.4% of SC PhD students).
- Women head 11.24% of 1,352 institutions and about 8.5% of INIs (14 of 164); no IIT is headed by a woman. The pool of women professors (31.3%) is larger than the posts they hold.

SC, ST and OBC

- At the IISERs, SC, ST and OBC students are 47.6% (4,418 of 9,276) against the 49.5% reserved share, with Pune falling steadily (−0.8 points a year) and PG lower than UG at all five IISERs that report both (41.6% against 49.2%). Berhampur's latest figure is probably a reporting change.
- Nationally the SC share falls from 15.6% at UG to 10.0% at the PhD and ST from 6.3% to 4.1%, mostly from PG to PhD. At INIs faculty are 26.1% from these groups against 46.3% of students (ratio 0.56) while central universities show none of that gap.
- At IISER Kolkata faculty are 21 of 131 (16.0%) from these groups against 47.0% of students: SC 6.1%, ST 0.8%, OBC 9.2%, each below the INI average. All seven appointments of the latest cycle were from reserved categories, which moved the share from 11.3% to 16.0%.
- The combined number can hide the groups that matter most: OBC supplies 27 of the 49.5 reserved points, so an institute can reach the line through OBC students alone. NIRF cannot show SC and ST separately.

Performance and choice of department

- Women are not weaker. Mean CGPA is level or ahead later (semester 10: +0.14 on average; +0.04 without two batches), and mixed in the first two years. By subject, women are ahead in humanities (+0.16), biology (+0.25) and earth sciences (+0.24) and behind in physics (−0.18), mathematics (−0.15) and computer science (−0.12), by small amounts that vary by batch. **The gap is too small to be meaningful.**
- Departments differ far more than batches: physics has lost about half its share of women (23% to 12%), biology has grown to about half (39% to 50%). Across subjects, where women do relatively worse is where women are fewer (rank correlation 0.90, exact p = 0.08); across whole batches, more men per woman goes with a larger lead for women (0.86 at semester 4, p = 0.007); within a department it runs the other way (−0.16 per 10 points). These are small gaps, but they show a trend; the data cannot say why.

Rank and preference

- Women's share among five-year students rises as an institute sits lower in the NIRF order: Spearman 0.82 across the seven IISERs (exact p = 0.03), 0.86 for the preference order (p = 0.02), 0.88 with IISc added (p = 0.007); none for faculty. **It is a pattern at institute level, not proof of the mechanism.**

Before IISER

- The women in the nearby settlements and the women at IISERs are **two different sample sets; we do NOT claim** that we are doing a correlation study. We simply noticed a pattern amongst the female students near us and wondered if similar patterns would arise in other places. It was worth a try at least. We started by looking at a rural settlement because, given that a large population of India lives in villages, the reasons we observe here might show us a clearer picture of what the women in other regions of the country face. However, this was a small convenience sample (not a random one), and the **findings should be treated with caution**. **We only state what we observed; we do not claim that similar patterns arise in other places.**

- Among 90 women in two settlements next to campus, 25% never attended school; marriage (59%) and cost (55%) were the most common reasons for stopping; the decision was usually someone else's; younger women have studied much further than older women; 43 of 77 would like to learn something now, mostly a practical skill; and the wish that girls study far is nearly universal. It is a convenience sample and does not describe who reaches IISER.

Limitations: What the data cannot show

- Applicants, offers and joiners by gender, category and route; PhD students and postdocs by gender and category at any IISER; faculty by category except at IISER Kolkata; students' states of origin; field of faculty at other institutes; SC and ST separately; non-binary people; anything below director level in leadership; and why.

6.1 The reasons in one place

A summary of every explanation we received, whether we could test it, and what we found. IISERs are government science institutes that admit students after Class 12 through one national test, the IAT, by rank. Women are 27.2% of undergraduates at IISER Kolkata, 14.4% of its faculty, and very few of its faculty are from SC/ST groups.

Fewer women students

Explanation	Provided by	Status and what we found
Rank in the national test	Dr. Anuradha Bhat Dr. Rumi De	**Tested. Supported as a pattern in the data set we observed, not conclusive due to the small sample size. We do NOT draw any conclusions from this.** No seat reservation for women, so seats go by rank; the most preferred institutes (Pune, IISER Kolkata) have the fewest women (about 27%) and the least preferred the most (about 52%): $\rho = 0.82$ across the seven IISERs, 0.88 with IISc. The likely reason, from the interviews, is that the top ranks are dominated by boys, because coaching and family investment favour sons. The mechanism could not be tested: applicants and ranks by gender are not public.
Home state	Dr. Anuradha Bhat Dr. Anindita Bhadra Dr. Koel Das Dr. Rumi De	**Tested. Not supported.** IISER Kolkata and Thiruvananthapuram take similar shares from their own state yet differ by 24 points in women.
Women differ from men in "capability"	Not provided by anyone. Dr. Bhattacharyya (DoAA) said this is a common perception. He provided us with the data to test this.	**Tested. Not supported.** Once admitted, women's grades are either level with or better than men's; however, the difference in grade points is too small to draw any conclusions.

Explanation	Provided by	Status and what we found
How girls are raised	All: Dr. Rahul Banerjee Dr. Koel Das Dr. Rumi De Dr. Neelanjana Sengupta Dr. Anindita Bhadra Dr. Anuradha Bhat Dr. Malancha Ta Dr. Rangeet Bhattacharyya	**Not tested.** Boys are encouraged more than girls, and girls meet prejudice about studying far from home. Some families may move boys to private schools and keep girls in government ones; boys are pushed toward engineering and medicine for quick jobs; some families are uncomfortable with a daughter being the only girl in a class. Children asked to draw a scientist or engineer mostly draw men (Dr. Bhadra, Dr. Bhat).
Location and safety	All	**Not tested.** Parents may hesitate over a five-year residential course right after Class 12; the south is seen as safer (as said by Dr. Bhadra and Dr. De), so Thiruvananthapuram and Tirupati attract more girls; students from Kerala used to be a large group at IISER Kolkata, but their numbers fell once two IISERs opened in the south (not testable; we do not have public data to support this); parents may prefer an institute in a nearby state (not checkable, because students' states of origin are not public).
Hostels	Dr. Rahul Banerjee Dr. Koel Das Dr. Rumi De	**Not tested.** Undergraduates all get hostels, but female PhD students at IISER Kolkata did not, which may deter them.
Role models	Dr. Neelanjana Sengupta Dr. Koel Das	**Not tested.** Interviewees disagreed on how much a lack of women teachers matters. About half of our small student survey said a great deal; 38% said it does not matter. Dr. Ta thinks women in positions of decision-making power matter more.
Biased exams and over-competitive entry tests	Dr. Rangeet Bhattacharyya Dr. Malancha Ta Dr. Rumi De	**Not tested.** The same question framed differently can give boys and girls different marks (a study cited by Prof. Godbole). Women are a minority in both entrance exams, JEE and NEET, and the gap is wider in JEE (we did not quantify this).

Fewer women faculty (14.4% at IISER Kolkata, flat since 2019; 10.0% at IISc)

Explanation	Provided by	Status and what we found
Small faculty makes percentages unreliable	Dr. Anuradha Bhat Dr. Anindita Bhadra	**Tested. Not conclusive.** IISER Kolkata has the third-largest faculty of the seven IISERs (Report 4).

Explanation	Provided by	Status and what we found
Two-body problem	All: Dr. Rahul Banerjee Dr. Koel Das Dr. Rumi De Dr. Neelanjana Sengupta Dr. Anindita Bhadra Dr. Anuradha Bhat Dr. Malancha Ta Dr. Rangeet Bhattacharyya	**Not tested.** Women often compromise on their career when a couple needs to live in the same place; most institutes in India do not hire both spouses, and there are only a few other employment opportunities near IISER Kolkata. People might need to commute to the main city of Kolkata to find jobs.
Leaky pipeline	All	**Not tested.** Numbers fall at graduation, PhD, postdoc and faculty; the path is long and coincides with marriage and childcare; no credit for a career break when hiring, and re-entry grants (about three years) are too short. Women are 27% of undergraduates but 14% of faculty, and intakes in some subjects are small (one year, two women among 50 physics students); 14% is hard to explain even so.
Few women apply	DoFA (Dr. Rahul Banerjee)	**Not tested.** There is a lack of female applicants. According to the DoFA, in one Chemistry search about 20 of nearly 200 applicants were women, and in some areas one or none. Applicant data were not stored.
Slow hiring and senior-heavy faculty	DoFA Dr. Koel Das	**Consistent with the data, not proven.** Few appointments between about 2012 and 2019, while Pune hired faster; of 40+ recruits since 2013 about 8 were women. Most women faculty joined in 2010-12 and are now senior; about 87 of roughly 130 faculty are full professors.
Remote location and safety	All	**Not tested.** The campus is far from the city, which makes commuting and family life harder; the DoFA also raised the perceived safety of Kolkata as a city.
Workplace attitudes and leadership	Dr. Koel Das Dr. Neelanjana Sengupta Dr. Malancha Ta Dr. Anindita Bhadra	**Not tested.** Condescension, being invited as “the woman on the committee”, and comments about family and appearance even at senior levels. About 11% of directors and deans across India are women against about 25% of faculty.
Institutional support and isolation	Dr. Rumi De	**Not tested.** A lone woman in a department is more likely to leave.

Fewer women in physics and maths than in biology and chemistry

Physics has lost about half its share of women (23% to 12%), while biology has grown to about half.

Explanation	Provided byProvided by	Status and what we found
Grades and department choice	Dr. Rangeet Bhattacharyya suggested that we look into this angle	**Not conclusive.** Women do relatively worse in subjects where women are fewer (correlation 0.90, exact p = 0.08), but with so little data we cannot say more; grades alone do not explain the ratios, and the grade pattern may itself come from the social factors below.
Women choose biology themselves; biology is broad and welcoming	Dr. Neelanjana Sengupta	**Not tested.** People from many disciplines work in biology.
"Easy" and "hard" sciences	Dr. Neelanjana Sengupta Dr. Anindita Bhadra Dr. Koel Das Dr. Rumi De	**Not tested.** Biology and chemistry are seen as easier and physics and maths as harder; from a young age girls are discouraged from ambition and from the "hard" sciences, and even educated parents may steer a girl to biology. The pattern shows in school (one technical-drawing class of 43 had 3 girls). Women who do choose maths do very well.
Jobs and time	Dr. Koel Das Dr. Rumi De	**Not tested.** Physics and maths may be seen as offering fewer jobs and taking longer to complete.
Nerd stigma	Dr. Koel Das Dr. Anindita Bhadra	**Not tested.** In the US, studies found girls fear being called "nerds"; two faculty doubted this applies in India. The Indian parallel may be families discouraging daughters from being too ambitious.
Earth Sciences puzzle	Dr. Rangeet Bhattacharyya	**Open.** It also has few women but is not maths-heavy, so "maths is the barrier" cannot be the full answer (DoAA).
Starts early	All	The issue starts well before IISER, which is why it shows in students, departments and faculty alike. Physics has one woman among about 30 faculty and maths one among 19; the latest hiring round attracted few physics candidates.

Fewer SC/ST faculty (IISER Kolkata: SC 6.1%, ST 0.8%, against about 47% reserved-category students)

Explanation	Provided by	Status and what we found
Quota not mandated before 2019	Dr. Banerjee (DoFA) Dr. Bhattacharyya (DoAA) Dr. Koel Das	**Confirmed by both deans and Dr. Koel Das.** Student reservation has long been mandated; for faculty the quota was only recommended before 2019. After 2019 a reserved post must stay vacant if no suitable candidate is found, and new recruitment stalls unless special drives are held. Counts have risen across India since, including at IISc, and all seven of IISER Kolkata's latest appointments were from reserved categories.

Explanation	Provided by	Status and what we found
Social causes	DoFA	**Not tested.** The DoFA said the initial low numbers reflect deeply rooted social issues that other studies have covered. Historical marginalisation is a hypothesis only; we could not interview the SC/ST Cell, and applicant data by category are not available, so we make no claim.

Suggestions we received from the experts during the interviews

- **Hostels for female PhD students (they are currently under construction)**
- **A visible policy of hiring spouses together.**
- **More women in decision-making.**
- **Safe transport for students**
- **More residential facilities on campus. A crèche and campus school already exist and have helped women faculty with small children. More such facilities would be useful.**
- **Awards for top women students (opinions on this differed)**

7. Conclusion

The case study set out to see who is missing from IISER Kolkata, and ended by finding that the answer begins well before it. At IISER Kolkata, women are about a quarter of undergraduates and, in the NIRF 2026 list, 14.4% of regular faculty (about 16% if the newest appointments are counted). That share has been flat for years, and it is low for the Institutes of National Importance as a whole, not only for us. SC, ST and OBC students are close to the reserved share across the IISERs, but faculty at IISER Kolkata are far below it, as they are across INIs, and the national fall for SC and ST students is steepest between postgraduate study and the PhD.

We tested the obvious explanations. Women do not do worse once admitted. Home state does not explain it. Faculty size does not explain it. What does show up is a pattern: the institutes most students prefer, which are those that attract the highest entrance ranks, have the fewest women (Spearman 0.82 across the seven IISERs, 0.88 with IISc), and the departments with the fewest women are the ones where women's relative grades are lowest. We emphasise that this is **not** due to the perceived "incapability" but rather due to various societal factors operating simultaneously (Dr. Bhattacharyya). Dr. De and Dr. Bhat also clarified that this shows a difference in societal expectations and the way women are conditioned from a young age, family investments in education and support; they do **NOT** show a capability difference between the sexes. Eight interviews and a village survey all point to the same kind of cause: who reaches the top ranks, family investment and marriage, the two-body problem and location, and the thin presence of women in leadership.

What we could not do is as important. Our sample size has been too small and inconsistent to reach any significant conclusions. Without more data, we cannot say which of the above causes carries weight. We can say where to look and what to ask the institutes to count.